Geometry and Topology-Guided Protein Design

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1 What is the problem that you will be investigating? Why is it interesting?

This project will explore geometric and topological property-guided protein design, specifically, gaussian curvature and mathematical knot type. Property guided protein design attracted attention from the computational biology community due to its potential to create novel proteins with desired functions [1, 2, 3]. Proteins with specific geometric and topological properties have been shown to exhibit unique stability and functionality characteristics [4, 5, 6]. For example, proteins with aligned gaussian curvature regions may have enhanced binding capabilities [7], while knotted proteins can exhibit increased stability and resistance to degradation [8, 9, 10]. By leveraging these properties, we can design proteins that are better suited for specific applications, such as drug delivery, enzyme catalysis, and material science. Previously, chemists have attempted to design knotted proteins experimentally [11, 12]. These efforts have provided valuable insights into the folding mechanisms and stability of knotted proteins, but they have also highlighted the challenges associated with designing such complex structures and chemical synthesis. This project aims to develop a computational protein design framework that integrates geometric and topological properties into the ESM3 [13] using derivative free guidance [14],leading to more effective and efficient design strategies.

2 What dataset(s) are you using? Is that sufficient?

For gaussian curvature-guided protein design, I will use the PDBBind v2020 dataset [15], which contains experimentally measured binding affinities for protein-ligand complexes along with their 3D structures. For knot type-guided protein design, I will use the KnotProt database [16], which contains a curated collection of knotted proteins along with their structural and functional annotations. Currently, both datasets appear to be sufficient for the scope of the project as they provide the necessary structural information and annotations.

3 What geometric/topological approach will you use/develop?

For gaussian curvature-guided protein design, I will compute the per face and per vertex gaussian curvature of the protein surface mesh. Detailed steps include:

1. Generate a surface mesh of the protein using the MSMS algorithm [17] from the 3D atomic coordinates.
2. Calculate the gaussian curvature at each vertex using the second fundamental form of the surface [18].

*GitHub repository for the project can be found at `https://github.com/annasu1225/Molecular_Topology`.

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For knot type-guided protein design, I will determine the knot type of the protein backbone using the Alexander polynomial invariant [19, 20]. Detailed steps include:

1. Extract the protein backbone coordinates from the 3D structure.
2. Simplify the backbone to a polygonal chain and project it onto a plane to obtain a knot diagram.
3. Compute the Alexander polynomial from the knot diagram to classify the knot type.

The above properties will be integrated into ESM3 using derivative-free guidance [14].

4 What are anticipated challenges of this project?

Challenge 1: The computational complexity of calculating geometric and topological properties for large protein structures. To address this, I will explore efficient algorithms and parallel computing techniques to speed up the calculations.

Challenge 2: Ensuring that the integration of geometric and topological properties into the protein design process does not compromise the overall stability and functionality of the designed proteins. To mitigate this, I will implement a multi-objective optimization approach that balances geometric and topological properties with traditional design criteria.

5 How will you evaluate your results?

5.1 Qualitatively, what kind of results do you expect (e.g. plots or figures)?

1. Visualizations of designed protein structures with highlighted regions of high and low gaussian curvature.
2. Diagrams and videos showing the knot types of designed proteins, along with their corresponding Alexander polynomials.

5.2 Quantitatively, what kind of analysis will you use to evaluate and/or compare your results (e.g. what performance metrics or statistical tests)?

1. Statistics on the distribution of stability of designed knotted proteins compared to unknotted proteins.
2. Correlation analysis between geometric/topological properties and binding affinities to identify potential design principles.

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